

**Intrusive and Voluntary Memory from a Single System: A Retrieved-Context Account of
Selective Interference**

Jordan B Gunn

Department of Psychology, University of Cambridge

Author Note

This is an in-progress working draft. Please do not cite, quote, or circulate without the authors' written permission.

Abstract

Visuospatial tasks performed after trauma-film encoding, or after a later reminder of the film, can reduce intrusive memories while leaving voluntary memory relatively intact. This selective interference effect is often treated as evidence that intrusive and voluntary remembering depend on differently vulnerable memory representations. Here we simulate an alternative retrieved-context account in an emotional Context Maintenance and Retrieval (eCMR) model, in which context reinstatement, competitor encoding, and retrieval control produce the selective-interference pattern without separately vulnerable representations. In the model, film context remains active after encoding and can later be reinstated by reminders, so an interfering task performed in either state associates task items with context states similar to film context. When later retrieval contexts cue both film items and task competitors, film items are less likely to be sampled during unguided retrieval, reducing intrusions even though film-item strength is not reduced. Deliberate recall is less vulnerable when task goals provide a film-directed start cue and selection favors film items over competitors; externally supplied film-related cues can also steer access toward film context across test formats. Together, the simulations reproduce selective impairment of unguided retrieval relative to deliberate recall and specify when the intrusion-voluntary-memory dissociation should be more or less pronounced, helping to interpret heterogeneity across empirical paradigms. Selective interference is therefore compatible with a single retrieved-context system and is not by itself diagnostic of separate representational systems or selective trace weakening.

Keywords: intrusive memories, voluntary memory, trauma-film paradigm, visuospatial intervention, selective interference, retrieved-context theory, computational modeling

Intrusive and Voluntary Memory from a Single System: A Retrieved-Context Account of Selective Interference

Introduction

Traumatic experience foregrounds a tension between durability and selectivity in episodic memory. Highly consequential events should be retained well enough in memory that critical details remain accessible long into the future. At the same time, adaptive retrieval is selective: what comes to mind should normally depend on current cues, goals, and safety. Intrusive memories after trauma suggest that durable encoding can outpace regulatory control. The event returns without being deliberately sought and without being well matched to the demands of the present ([Brewin et al., 2010](#); [Ehlers & Clark, 2000](#); [Iyadurai et al., 2019](#); [Krans et al., 2009](#)). Understanding why such memories arise, and how they differ from voluntary memory for the same event, is therefore a shared target for both clinical science and basic memory theory.

The trauma-film paradigm provides a controlled setting for comparing intrusive and voluntary memory for the same event ([Holmes & Bourne, 2008](#); [James et al., 2016](#)). Participants view distressing film material, record later intrusive memories of the film, and often complete voluntary-memory tests for the same material. This design makes it possible to ask whether the same encoded event can remain available to deliberate retrieval while becoming more or less likely to return involuntarily. In such research, voluntary memory can be relatively preserved, and its relationship to intrusion frequency can be weak, such that the two measures dissociate ([Holmes et al., 2004](#); [James et al., 2016](#)). The same encoded episode can therefore support both involuntary and deliberate access, and these modes can come apart empirically ([Holmes et al., 2004](#); [Iyadurai et al., 2019](#)).

Visuospatial interference tasks make this dissociation experimentally sharper. When brief tasks such as Tetris are performed after film encoding, subsequent intrusions can be reduced while voluntary memory for the same material shows little measurable impairment ([Deepröse et al., 2012](#); [Holmes et al., 2009, 2010](#); [Lau-Zhu et al., 2019](#)). Related post-reminder studies show that a reminder-plus-Tetris procedure can reduce later intrusions after a longer delay ([James et al., 2015](#);

[Kessler et al., 2020](#)). Similar procedures have also been extended to real-life traumatic events, including motor vehicle accidents, emergency caesarean section, and war-related trauma ([Horsch et al., 2017](#); [Iyadurai et al., 2018](#); [Kanstrup et al., 2021](#)). At the same time, the evidence base is heterogeneous, and recent reviews and replications caution against treating the effect as settled or uniform ([Asselbergs et al., 2023](#); [Varma et al., 2024](#); [Wessel et al., 2025](#)). This selective interference effect therefore raises a theoretical question: when and why can a post-encoding or post-reminder manipulation alter one mode of retrieval more than another, even while both draw on the same recent experience?

One prominent family of interpretations treats the selective interference effect as evidence that the memory underlying intrusions is more vulnerable to interference than the memory supporting deliberate recall ([Brewin et al., 1996](#); [Brewin, 2014](#)). Dual-representation accounts make this claim architecturally: traumatic experience is proposed to produce distinct sensory and contextual memory traces, with the first supporting involuntary re-experiencing and the second supporting deliberate recall ([Brewin et al., 2010](#); [Brewin, 2014](#)). In immediate post-film interventions, a visuospatial task is then understood to compete with consolidation of the sensory or intrusion-supporting trace, reducing later intrusions while leaving the contextual trace and voluntary memory relatively preserved ([Deepröse et al., 2012](#); [Holmes et al., 2009, 2010](#)). For post-reminder interventions, related interpretations often appeal to reconsolidation or post-reactivation restabilization: the reminder reactivates the relevant trace, and the interfering task disrupts its later availability ([James et al., 2015](#); [Kindt et al., 2009](#)). Across these variants, selective interference is explained by treating the manipulation as acting more strongly on intrusion-supporting or reactivated memory than on the basis for deliberate recall.

Here we argue that selective interference need not be interpreted as evidence for differently vulnerable memory representations. It may instead reflect differences in retrieval processes operating over a shared set of episodic representations. Mechanisms supporting selective, goal-directed retrieval in voluntary recall may be sufficient to sidestep or overcome interference that nonetheless reduces the likelihood of unintentional recall. On this interpretation,

the selective-interference effect reflects a capacity to intentionally focus retrieval on goal-relevant event information while reducing the influence of competing memories.

We formalize and evaluate this proposal in an emotional Context Maintenance and Retrieval model (Cohen & Kahana, 2022; Polyn et al., 2009a; Talmi et al., 2019).

Retrieved-context models explain recall by treating context as a gradually changing state that is updated by studied items, reinstated by recalled items, and used to cue later memory search (Howard & Kahana, 2002; Polyn et al., 2009a; Sederberg et al., 2008). CMR extends this framework by representing context-to-item and item-to-context associations across temporal, semantic, and source/contextual dimensions (Polyn et al., 2009a, 2009b). Control over memory search enters the framework in two related ways: start-of-recall context reinstatement can bias retrieval toward a target episode, while monitoring mechanisms can reject candidates that poorly match a target context (Lohnas et al., 2015; Sederberg et al., 2008). Emotional CMR variants (i.e., eCMR and CMR3) add an affective-context stream and emotion-dependent binding, allowing retrieved-context models to capture how affective content shapes recall organization, emotional intrusions, and intentional access (Cohen & Kahana, 2022; Talmi et al., 2019). Together, these commitments make the framework a principled setting for testing whether selective interference can arise within a single affectively modulated episodic-memory system.

For selective interference, our model treats the film, any reminder, and subsequent interference as episodes encoded by binding item representations to a continuously evolving context representation (Howard & Kahana, 2002; Polyn et al., 2009a; Yonelinas et al., 2019). Post-film or post-reminder activity can therefore encode task items in context states similar to film-associated context states, including states reinstated by reminders. At retrieval, the current context state cues memory search, so competition depends on how strongly film items and task items are associated with similar context states. Deliberate recall can reduce vulnerability to this competition when retrieval begins from a more film-directed state and when selection favors film items over competitors. The account therefore attributes the intrusion-voluntary-memory dissociation to retrieval modes and control settings applied to shared memory associations. This

reframes selective interference as a change in which competitors are cued by film-associated context states, rather than as reduced availability of film items after encoding or reminder reactivation.

Our simulations evaluate the account at the level where a retrieved-context model makes its distinctive predictions: the distribution and organization of recall across the film, interference, and delay phases of a trauma-film paradigm. They first demonstrate the core selective-interference dissociation without separately vulnerable representations: an interference task can reduce unguided retrieval more than deliberate recall, even though the same stored film episode supports both. They then distinguish the control mechanisms that protect deliberate recall, showing how a more film-directed retrieval state and a sharper decision rule produce different patterns of impairment in both recall rates and recall organization. The next simulations separate the contributions of task encoding strength and contextual overlap, clarifying when task items are cued alongside film items and when they become strong competitors for film retrieval. They also apply the same overlap logic to delayed interference, asking whether reminder-driven reinstatement makes later task items competitors by returning encoding context toward the film. Follow-up simulations use the same framework to interpret heterogeneity across empirical paradigms, including cue-embedded intrusion and recognition tasks ([Lau-Zhu et al., 2021](#)). They show that film-related cues during intrusion monitoring can narrow the intrusion-voluntary-memory dissociation by steering retrieval toward film context. They also show that tests supplying a film item can be less vulnerable to interference, because the supplied item can reinstate its associated context instead of requiring retrieval to sample among film items and task competitors from a broad contextual cue. The result is a mechanistic alternative to separately vulnerable memory representations and a set of specific hypotheses about when selective interference should appear and how it should vary across paradigms.

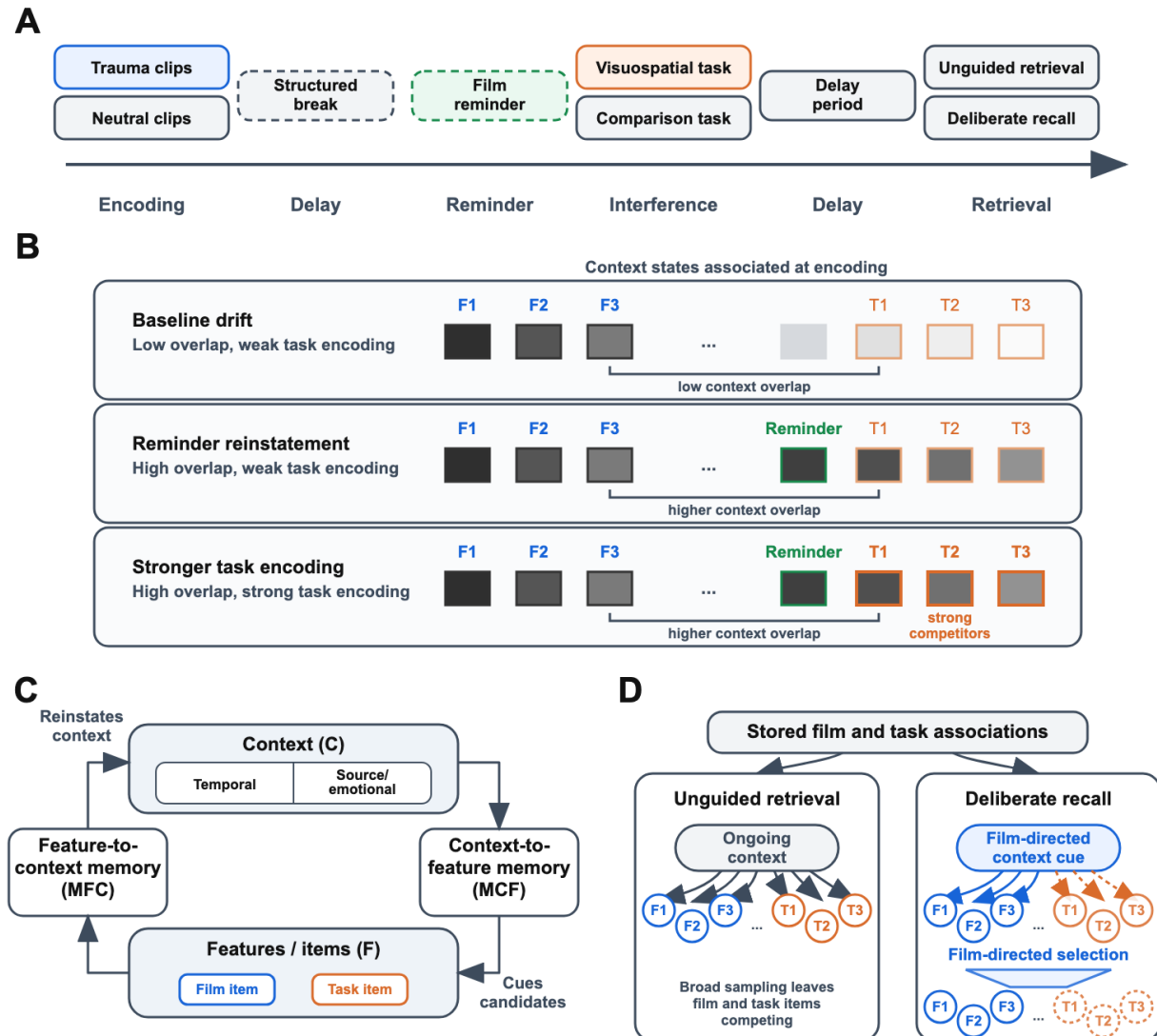
A Retrieved-Context Account of Selective Interference

The simulations rely on three retrieved-context mechanisms. Context-to-item retrieval uses current context to cue candidate items. Competition follows when film items and task

competitors are associated with similar context states. Item-to-context retrieval allows recalled items, reminder cues, and supplied film items to reinstate associated context. Film-directed retrieval control changes the starting cue and selection rule used to favor film items over task competitors. Figure 1 shows how these mechanisms map onto the trauma-film paradigm.

Figure 1

Retrieved-context account of selective interference. The account explains selective interference through context overlap, task encoding strength, and retrieval control within a shared memory system. **(A)** The simulated paradigm includes film encoding, a structured break, a film reminder, interference or comparison task, a delay period, and retrieval tests. **(B)** Film and task items remain distinct while becoming associated with context states active at encoding; contextual overlap makes task items likely to be cued alongside film items, and task encoding strength determines how strongly they compete once cued. **(C)** Context-to-item associations cue candidate items, and item-to-context associations let recalled, reminded, or supplied items reinstate associated context. **(D)** The same stored associations support unguided retrieval and deliberate recall, with deliberate recall gaining protection from a film-directed start cue and film-directed selection.



Contextual similarity is established during encoding. Film items and task items are distinct item representations, but each is linked to the context state active when it is encountered. Task items are most likely to share film-relevant context when they are encoded soon after the film, after a reminder has reinstated film-associated context, or under matching affective conditions. Delay and filler activity move context away from the film, reducing the likelihood that later retrieval cues film items and task competitors together. These conditions determine whether task items are likely to be cued alongside film items; task encoding strength determines how strongly they compete once cued.

Context-to-item retrieval uses the current context state to cue candidate items. Here, unguided retrieval refers to broad context-to-item sampling with little film-directed guidance from task goals or external cues. When a retrieval context resembles states associated with both film items and task items, the same cue supports both sets. Such broad, weakly guided retrieval is especially vulnerable to this competition: as task competitors receive support, any given film item becomes less likely to be sampled as an intrusion. The size of this effect depends on the number of competitors, their learned support, and the similarity between their associated temporal or affective context states and film context. Perceptual modality contributes through those quantities: a visuospatial task is effective to the extent that it creates competitors with sufficient learned support or sufficiently similar associated context.

Item-to-context associations allow recalled items, reminder cues, and supplied film items to reinstate associated context. In ordinary recall, a retrieved item shifts context toward the state in which that item was encoded, making nearby studied items more likely to be retrieved next. This context-updating loop explains temporal contiguity and temporally organized recall across filled delays ([Howard & Kahana, 2002](#); [Kahana, 1996](#); [Sederberg et al., 2008](#)). The same dynamics allow recent but irrelevant experiences to intrude during recall of a target episode, with intrusion frequency decaying gradually over intervening time ([Howard et al., 2008](#); [Lohnas et al., 2015](#); [Zaromb et al., 2006](#)). In delayed-interference paradigms, a reminder of the film is used before task encoding in studies that interpret the procedure as memory reactivation ([James et al., 2015](#);

[Kessler et al., 2020](#)). On the present account, the reminder uses the same item-to-context pathway to reinstate film-associated context, paralleling temporal-context explanations of human post-reminder effects ([Sederberg et al., 2011](#)). Delayed interference is therefore graded by reminder strength and subsequent drift, because those quantities determine how much film-associated context remains available when the task is encoded.

The same item-to-context pathway also matters at test. During recall or monitoring, retrieved or presented items update context as retrieval unfolds. Retrieving a film item pulls context toward film-associated states and can support additional film retrieval, whereas retrieving a task competitor pulls context toward task-associated states and can prolong competition. Film-related cues supplied during test can likewise update context, making subsequent retrieval less dependent on internal cue maintenance. Furthermore, retrieval tests that supply a film item can be less vulnerable to interference, because the supplied item can reinstate its associated context directly, sidestepping the problem of selecting between items associated with overlapping context states ([Lau-Zhu et al., 2021](#)).

Deliberate recall is protected only to the extent that retrieval control changes the search. Deliberate recall using context-to-item retrieval remains exposed to competitor interference. The first control lever is cue construction or start-state control, which reinstates film-associated context before recall and biases search toward film items. The second is competition focusing or filtering, which sharpens selection among candidates and favors strongly supported film items over weaker or off-target competitors. These levers yield graded protection: deliberate recall can lose film items when competition is strong, but endogenous film-directed control can reduce that loss relative to broad retrieval from ongoing context.

Emotional extensions of CMR represent affect as a source-like context dimension ([Cohen & Kahana, 2022](#); [Talmi et al., 2019](#)). In the present model, this affective-context stream can amplify competition when film and task items share the relevant affective feature. The formal model specified next instantiates the central commitments as phase-specific encoding, reminder-driven reinstatement, retrieval-mode and control settings, and affective overlap.

Model and Paradigm

The simulations instantiate the retrieved-context account in an emotional CMR model (Cohen & Kahana, 2022; Talmi et al., 2019). The model has a feature layer, a context layer, feature-to-context memory M^{FC} , and context-to-feature memory M^{CF} (Polyn et al., 2009a). It updates context, learns item-context associations by outer products, retrieves items competitively, and reinstates context after retrieval. The selective-interference simulations add two paradigm-specific commitments. First, film, break/delay, post-film task, and filler/delay periods form a continuous encoded stream, and a film reminder can reinstate film-associated context before task encoding. Second, non-film events from the break, task, and filler phases are encoded as distinct recallable items that can compete with film items during retrieval. This setup lets selective interference appear as a redistribution of retrieval across encoded phases, rather than as a direct reduction in film-item strength. The equations below specify the feature, context, and memory representations that implement this process.

Feature, Context, and Memory Representations

All vectors are column vectors. Each study event i has an orthonormal feature vector $f_i \in \mathbb{R}^N$, where N is the number of encoded item features in the simulated paradigm. The model maintains a unit-length context vector $c_i \in \mathbb{R}^D$, where D is the number of context units. The primary context units represent the start state and item-associated temporal context. Affective-overlap variants add a two-unit vector s_i for each item, with $s_i = [1, 0]^\top$ for items assigned the marked affective category and $s_i = [0, 1]^\top$ for neutral or default items. These category units are used only in the sensitivity machinery described below; the core encoding, reminder, and retrieval operations are written over the full context vector.

Feature-to-context memory maps item features to context. The matrix $M^{FC} \in \mathbb{R}^{D \times N}$ maps feature vectors to context input. Context-to-feature memory maps context to item support. The matrix $M^{CF} \in \mathbb{R}^{N \times D}$ maps context to feature support.

Table 1 summarizes the model structures.

Table 1

Model structures and dimensions. The column-vector convention makes $M^{FC}f$ a context vector and $M^{CF}c$ a feature-support vector.

Symbol	Name	Dimension	Role
f_i	feature vector	$N \times 1$	Orthonormal item representation
c_i	context vector	$D \times 1$	Unit-length retrieval state
M^{FC}	feature-to-context memory	$D \times N$	Retrieves context from item features
M^{CF}	context-to-feature memory	$N \times D$	Retrieves item support from context
s_i	affective category vector	2×1	Optional category vector for affective variants

Pre-Experimental Associations

Pre-experimental initialization establishes baseline associations before the experiment.

For the start unit and item-associated temporal context units:

$$M_{pre}^{FC}(j, i) = \begin{cases} 1 - \gamma & \text{if } j = i + 1 \\ 0 & \text{otherwise} \end{cases}$$

$$M_{pre}^{CF}(i, j) = \begin{cases} 0 & \text{if } j = 0 \\ \delta & \text{if } j = i + 1 \\ \alpha & \text{if } j > 0 \text{ and } j \neq i + 1 \end{cases}$$

The start context has no pre-experimental context-to-feature support. For context units

corresponding to items, δ gives each item self-support and α gives all other items shared support. This initialization supplies baseline support before experimental learning and is treated as fixed unless a simulation explicitly varies it. In affective-category variants, the category rows of M^{FC} are initialized so that items cue their assigned affective category, while affective context-to-feature support begins at zero. This lets affective category update context without contributing retrieval competition unless affective context-to-feature learning is enabled.

Encoding Across Phases

The film, break, task, and filler phases use the same encoding operation with phase-specific rates. When a reminder is present, it uses the reinstatement operation described below before the task phase begins. Let p index the current encoding phase. Let β_p denote that phase's context drift rate and λ_p denote its context-to-feature learning multiplier. The film phase encodes target items with baseline rates. The break and filler phases add intervening experience. The task phase encodes post-film task items whose effect depends on their strength and the similarity between their associated context states and film context. Items encoded in any of these phases can enter the retrieval competition unless they have already been recalled.

When an item is presented, its active feature vector probes feature-to-context memory to retrieve context input:

$$c_i^{\text{IN}} = M^{FC} f_i$$

The model normalizes this input to unit length, yielding $\hat{c}_i^{\text{IN}}$. Context then drifts toward the normalized input:

$$c_i = \rho_i c_{i-1} + \beta_p \hat{c}_i^{\text{IN}}$$

$$\rho_i = \sqrt{1 + \beta_p^2 \left[(c_{i-1} \cdot \hat{c}_i^{\text{IN}})^2 - 1 \right]} - \beta_p (c_{i-1} \cdot \hat{c}_i^{\text{IN}})$$

The coefficient ρ_i normalizes the updated context vector, as in standard CMR

implementations. The drift rate β_p controls how far the current phase moves context away from the prior phase. In selective-interference simulations, this rate helps determine whether task items are encoded near or far from the film context.

After context updating, the active feature and updated context are bound by outer-product learning. Feature-to-context associations update as:

$$\Delta M^{FC} = \gamma c_i f_i^\top$$

A primacy function weights context-to-feature learning:

$$\phi_i = \phi_s e^{-\phi_d \cdot i} + 1$$

$$\Delta M^{CF} = \lambda_p \phi_i f_i c_i^\top$$

Here, λ_p is the phase-specific context-to-feature learning scale. Increasing λ_p during the task phase makes task items stronger competitors for later context-to-feature retrieval. Unless a later sensitivity analysis enables affective encoding-strength modulation, this learning remains primacy-driven.

Affective Category Context

The affective-category machinery represents two eCMR-style ingredients without making them core mechanisms of selective interference. First, affective category functions as a source-like context dimension: items assigned the same category can cue shared components of context. This lets the simulations treat a film and a task as affectively matched or mismatched without changing their item representations. Second, affective encoding-strength modulation can give items in the marked category stronger context-to-feature learning when that sensitivity analysis is enabled. Let $\phi_{\text{emot},i}$ denote the affective learning strength associated with item i . With the binary vector above, $\phi_{\text{emot},i} = \phi_{\text{emot}}[s_i]_1$, so items assigned the marked affective category receive ϕ_{emot} and neutral or default items receive 0. When enabled, this term scales context-to-feature learning for the affective

category components of context. The parameter ϕ_{emot} controls the learning strength assigned to the marked affective category. When $\phi_{\text{emot},i} = 0$, the item can still update affective category context but does not add affective context-to-feature support. When $\phi_{\text{emot}} = 0$, affective category context does not contribute retrieval competition. Simulation 3 uses this machinery to examine whether affective matching and affective encoding strength alter competitor effectiveness.

Reminder Reinstatement

Delayed-interference simulations insert a film reminder before the task phase. Before the reminder, context may drift toward the start state, capturing context change across the delay. The reminder cue then probes M^{FC} and updates current context using the same unit-length context-update rule defined above. No new associations are learned during this reminder step, and the reminder cue is not added as a recallable item. Its role is to reinstate film-associated context before ordinary task encoding resumes. Task items encoded after stronger reminder reinstatement are therefore associated with context states more similar to those associated with film items. Delayed interference is thus graded by reminder reinstatement and subsequent drift, rather than by an all-or-none reactivation window.

Retrieval, Control, and Termination

Context-to-feature retrieval underlies both unguided retrieval and deliberate recall. At the start of retrieval, context can drift toward its start state, with β_{start} controlling the strength of this reinstatement. The current context then cues encoded items through M^{CF} :

$$a_i = [M^{CF}c]_i$$

Because encoded items compete for retrieval, adding competitors in contextually similar states reduces each film item's share of retrieval probability. The choice sensitivity τ_c power-scales the combined activation vector:

$$A_i = a_i^{\tau_c}$$

The probability of recalling item i , conditional on continuing recall, then follows a Luce choice rule over currently unrecalled items:

$$P(i \mid \text{recall}) = \frac{A_i}{\sum_k A_k}$$

At each retrieval step, the model may terminate with probability $P_{\text{stop}}(j)$ or recall an item with probability $(1 - P_{\text{stop}}(j)) \cdot P(i \mid \text{recall})$. Termination probability grows with output position:

$$P_{\text{stop}}(j) = \theta_s e^{j\theta_r}$$

Here, j is the number of items already recalled, θ_s sets the baseline termination probability, and θ_r controls its growth. In simulations, this value is bounded to remain a valid probability. When an item is recalled, feature-to-context memory reinstates its associated context using the same unit-length context-update rule defined above. For target monitoring, let $t_i = 1$ for film items and $t_i = 0$ otherwise, and define $\kappa_i = t_i + (1 - t_i)\kappa$. When $\kappa < 1$, sampled non-film competitors have less influence on the next context state than sampled film items. This reinstatement drives context toward the recalled item’s encoding neighborhood, biasing subsequent retrievals toward nearby items. This recalled-item update produces temporal organization in retrieval.

The simulations vary retrieval mode and film-directed control settings without changing the stored film and task episodes. Unguided retrieval is modeled as relatively broad context-to-feature sampling from the current retrieval context. Deliberate recall uses the same context-to-feature pathway, but with stronger film-directed start reinstatement, sharper selection among candidates, and optionally reduced context updating after rejected non-film competitors. These settings do not create a separate voluntary-memory store; they change the retrieval state and selection policy applied to the same item-context associations.

Tests that supply a film item use feature-to-context retrieval to reduce reliance on broad context-to-feature search. When a test supplies a film item, that item can retrieve its own associated context through feature-to-context memory. This route supplies more target-specific

information than broad context-to-feature search. It should therefore be less exposed to competitors encoded near film-associated context, although recognition-like access is treated as a cue-driven case to be interpreted alongside the task design rather than as an absolute recognition result.

Parameters and Simulation Factors

The base parameter vector is anchored by external eCMR fits rather than fit to the selective-interference simulations themselves. For these simulations, the anchor is a pooled eCMR fit to the Ukraine war photographs free-recall data (Dupertuys et al., 2026). The selective-interference simulations then manipulate paradigm-level scale factors that determine phase drift, competitor learning, reminder reinstatement, and retrieval control. This separation keeps fitted memory-search parameters distinct from the theoretical manipulations used to create selective interference.

Table 2 lists the base parameters used by the model.

Each phase applies multiplicative scale factors to the base encoding rates. The film phase uses base rates directly. Subsequent phases modulate drift rate, which controls context proximity to preceding items, and context-to-feature learning, which controls competitor encoding strength.

This table defines β_p and λ_p for each encoded phase. The task-phase scales m_{task}^β and m_{task}^λ are central to the simulations. The drift scale controls how much context changes during task encoding. The context-to-feature learning scale operationalizes task encoding strength: better-supported competitors capture more retrieval probability from film targets once cued. Together, drift and learning scales define whether the task phase is merely added experience or a source of items that are both recruited into film-related context and strong competitors once cued.

Table 4 lists the broader simulation factors and their theoretical roles.

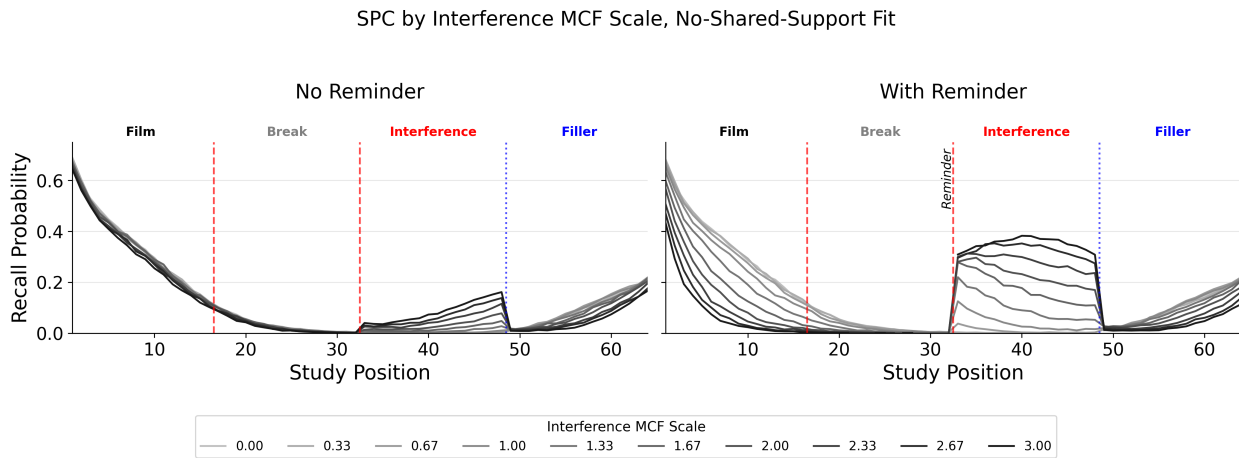
The simulation factors set the phase lengths, drift rates, learning scales, reminder strengths, and retrieval-control settings used in the sections below. Because retrieval proceeds as a sequence of competitive samples terminated by a growing stop probability, task competitors can reduce film retrieval by winning individual retrieval competitions before search terminates.

Simulations

The first rendered sweep included here orients the reader to the simulated phase structure before the later sections develop the final manuscript contrasts. These plots are diagnostic outputs from the interference-strength sweep, not stand-alone evidence for the full intrusion-voluntary-memory dissociation.

Figure 2

Phase-resolved recall in the interference-strength sweep. Serial-position curves show recall probability across the film, break, interference, and filler phases of the simulated paradigm, separately for no-reminder and reminder conditions. Darker lines indicate stronger context-to-item learning for interference-task items. The figure shows how the interference manipulation changes the distribution of retrieval across encoded phases, rather than only changing total film recall.



Simulation 1: Core Selective-Interference Dissociation

Premise: The basic selective-interference result is a 2 x 2 contrast between interference condition and retrieval mode. The section asks whether the same stored film episode and the same post-film interference manipulation can produce larger impairment in intrusion-like retrieval than in deliberate recall.

Intended text take-away: This section establishes the core sufficiency result. The model can reproduce the qualitative intrusion-voluntary-memory dissociation without separately

weakening film items for one measure. Deliberate recall may still be impaired; the claim is relative protection, not immunity.

Simulation 2: Control Mechanisms Underlying Deliberate-Recall Protection

Premise: Deliberate recall is not categorically protected. Its relative sparing depends on control operations that make retrieval more film-directed.

Intended text take-away: This section earns the “relatively intact voluntary memory” claim. Sparing is a graded consequence of retrieval control rather than a property attached to voluntary tests as a category. Weakening these control operations should make deliberate recall resemble intrusion-like retrieval more closely.

Simulation 3: Competitor Strength and Contextual Overlap

Premise: An interference task becomes effective when it encodes competitors with enough learned support and with associated context states similar enough to film context. This section asks what makes a task Tetris-like in model terms.

Intended text take-away: This section explains what makes interference effective. The account should not reduce to generic distraction, list length, or task difficulty. Competitor strength matters because competitors are retrievable from film-related context. The affective variant sharpens the boundary condition: stronger affective film encoding can increase baseline film retrieval and reduce vulnerability to mismatched task competitors, whereas affective/source matching can make strong task items more effective competitors.

Simulation 4: Reminder Reinstatement and Delayed Interference

Premise: The delayed-reminder case applies the overlap mechanism after context has drifted away from the film. A reminder matters because it reinstates film context before competitor encoding.

Intended text take-away: This section connects the model to post-reminder interventions. The reminder is not treated as a special storage-disruption device. It restores the contextual condition required for later competitor encoding to affect film-item retrieval.

Simulation 5: Heterogeneity Across Recall-Like Test Formats

Premise: Intrusion diaries, vigilance-style monitoring, free recall, and cued recall differ in how much they rely on broad context-to-item search versus film-directed retrieval support. This section maps the model's retrieval-control mechanisms onto variation across recall-like empirical measures.

Intended text take-away: This section interprets heterogeneity across empirical paradigms without treating intrusion and voluntary memory as fixed computational kinds. The model predicts a continuum of vulnerability across recall-like tests depending on how much the task supports film-directed retrieval.

Simulation 6: Probe-Driven Access in Recognition-Like Tests

Premise: Tests that supply a film item are a limiting case of voluntary access because they begin closer to the target item than broad recall does. This section preserves the useful part of the old recognition result without treating probe-driven access as categorically protected.

Intended text take-away: This section handles tests that supply a film item and recognition-like measures as a probe-driven access case. The claim is that probe-driven access reduces exposure to contextual competition, not that recognition escapes the mechanism in every design or that all voluntary memory has the same protection.

General Discussion**References**

- Asselbergs, J., van Bentum, J., Riper, H., Cuijpers, P., Holmes, E. A., & Sijbrandij, M. (2023). A systematic review and meta-analysis of the effect of cognitive interventions to prevent intrusive memories using the trauma film paradigm. *Journal of Psychiatric Research*, 159, 116–129. <https://doi.org/10.1016/j.jpsychires.2023.01.028>
- Brewin, C. R. (2014). Episodic memory, perceptual memory, and their interaction: Foundations for a theory of posttraumatic stress disorder. *Psychological Bulletin*, 140(1), 69–97. <https://doi.org/10.1037/a0033722>

- Brewin, C. R., Dalgleish, T., & Joseph, S. (1996). A dual representation theory of posttraumatic stress disorder. *Psychological Review*, 103(4), 670–686.
<https://doi.org/10.1037/0033-295X.103.4.670>
- Brewin, C. R., Gregory, J. D., Lipton, M., & Burgess, N. (2010). Intrusive images in psychological disorders: Characteristics, neural mechanisms, and treatment implications. *Psychological Review*, 117(1), 210–232. <https://doi.org/10.1037/a0018113>
- Cohen, R. T., & Kahana, M. J. (2022). A memory-based theory of emotional disorders. *Psychological Review*, 129(4), 742–776. <https://doi.org/10.1037/rev0000334>
- Deepröse, C., Zhang, S., DeJong, H., Dalgleish, T., & Holmes, E. A. (2012). Imagery in the aftermath of viewing a traumatic film: Using cognitive tasks to modulate the development of involuntary memory. *Journal of Behavior Therapy and Experimental Psychiatry*, 43(2), 758–764. <https://doi.org/10.1016/j.jbtep.2011.10.008>
- Dupertuys, J., Shulhan, A., & Talmi, D. (2026). *Ukraine Memory experiment data set*.
- Ehlers, A., & Clark, D. M. (2000). A cognitive model of posttraumatic stress disorder. *Behaviour Research and Therapy*, 38(4), 319–345. [https://doi.org/10.1016/S0005-7967\(99\)00123-0](https://doi.org/10.1016/S0005-7967(99)00123-0)
- Holmes, E. A., & Bourne, C. (2008). Inducing and modulating intrusive emotional memories: A review of the trauma film paradigm. *Acta Psychologica*, 127(3), 553–566.
<https://doi.org/10.1016/j.actpsy.2007.11.002>
- Holmes, E. A., Brewin, C. R., & Hennessey, R. G. (2004). Trauma films, information processing, and intrusive memory development. *Journal of Experimental Psychology: General*, 133(1), 3–22. <https://doi.org/10.1037/0096-3445.133.1.3>
- Holmes, E. A., James, E. L., Coode-Bate, T., & Deepröse, C. (2009). Can playing the computer game Tetris reduce the build-up of flashbacks for trauma? A proposal from cognitive science. *PLoS ONE*, 4(1), e4153. <https://doi.org/10.1371/journal.pone.0004153>
- Holmes, E. A., James, E. L., Kilford, E. J., & Deepröse, C. (2010). Key steps in developing a cognitive vaccine against traumatic flashbacks: Visuospatial Tetris versus verbal Pub Quiz. *PLoS ONE*, 5(11), e13706. <https://doi.org/10.1371/journal.pone.0013706>

- Horsch, A., Vial, Y., Favrod, C., Harari, M. M., Blackwell, S. E., Watson, P., Iyadurai, L., Bonsall, M. B., & Holmes, E. A. (2017). Reducing intrusive traumatic memories after emergency caesarean section: A proof-of-principle randomized controlled study. *Behaviour Research and Therapy*, 94, 36–47. <https://doi.org/10.1016/j.brat.2017.03.018>
- Howard, M. W., & Kahana, M. J. (2002). A distributed representation of temporal context. *Journal of Mathematical Psychology*, 46(3), 269–299. <https://doi.org/10.1006/jmps.2001.1388>
- Howard, M. W., Youker, T. E., & Venkatadass, V. S. (2008). The persistence of memory: Contiguity effects across hundreds of seconds. *Psychonomic Bulletin & Review*, 15(1), 58–63.
- Iyadurai, L., Blackwell, S. E., Meiser-Stedman, R., Watson, P. C., Bonsall, M. B., Geddes, J. R., Nobre, A. C., & Holmes, E. A. (2018). Preventing intrusive memories after trauma via a brief intervention involving Tetris computer game play in the emergency department: A proof-of-concept randomized controlled trial. *Molecular Psychiatry*, 23(3), 674–682. <https://doi.org/10.1038/mp.2017.23>
- Iyadurai, L., Visser, R. M., Lau-Zhu, A., Porcheret, K., Horsch, A., Holmes, E. A., & James, E. L. (2019). Intrusive memories of trauma: A target for research bridging cognitive science and its clinical application. *Clinical Psychology Review*, 69, 67–82. <https://doi.org/10.1016/j.cpr.2018.08.005>
- James, E. L., Bonsall, M. B., Hoppitt, L., Tunbridge, E. M., Geddes, J. R., Milton, A. L., & Holmes, E. A. (2015). Computer game play reduces intrusive memories of experimental trauma via reconsolidation-update mechanisms. *Psychological Science*, 26(8), 1201–1215. <https://doi.org/10.1177/0956797615583071>
- James, E. L., Lau-Zhu, A., Clark, I. A., Visser, R. M., Hageraars, M. A., & Holmes, E. A. (2016). The trauma film paradigm as an experimental psychopathology model of psychological trauma: Intrusive memories and beyond. *Clinical Psychology Review*, 47, 106–142. <https://doi.org/10.1016/j.cpr.2016.04.010>
- Kahana, M. J. (1996). Associative retrieval processes in free recall. *Memory & Cognition*, 24(1),

103–109.

Kanstrup, M., Kontio, E., Geranmayeh, A., Olofsdotter Lauri, K., Moulds, M. L., & Holmes, E.

A. (2021). A single case series using visuospatial task interference to reduce the number of visual intrusive memories of trauma with refugees. *Clinical Psychology & Psychotherapy*, 28(1), 109–123. <https://doi.org/10.1002/cpp.2489>

Kessler, H., Schmidt, A.-C., James, E. L., Blackwell, S. E., von Rauchhaupt, M., Harren, K.,

Kehyayan, A., Clark, I. A., Sauvage, M., Herpertz, S., Axmacher, N., & Holmes, E. A. (2020). Visuospatial computer game play after memory reminder delivered three days after a traumatic film reduces the number of intrusive memories of the experimental trauma. *Journal of Behavior Therapy and Experimental Psychiatry*, 67, 101454.

<https://doi.org/10.1016/j.jbtep.2019.01.006>

Kindt, M., Soeter, M., & Vervliet, B. (2009). Beyond extinction: Erasing human fear responses and preventing the return of fear. *Nature Neuroscience*, 12(3), 256–258.

<https://doi.org/10.1038/nn.2271>

Krans, J., Näring, G., Becker, E. S., & Holmes, E. A. (2009). Intrusive trauma memory: A review and functional analysis. *Applied Cognitive Psychology*, 23(8), 1076–1088.

<https://doi.org/10.1002/acp.1611>

Lau-Zhu, A., Henson, R. N., & Holmes, E. A. (2019). Intrusive memories and voluntary memory of a trauma film: Differential effects of a cognitive interference task after encoding. *Journal of Experimental Psychology: General*, 148(12), 2154–2180. <https://doi.org/10.1037/xge0000598>

Lau-Zhu, A., Henson, R. N., & Holmes, E. A. (2021). Selectively interfering with intrusive but not voluntary memories of a trauma film: Accounting for the role of associative memory.

Clinical Psychological Science, 9(6), 1128–1143. <https://doi.org/10.1177/2167702621998315>

Lohnas, L. J., Polyn, S. M., & Kahana, M. J. (2015). Expanding the scope of memory search: Modeling intralist and interlist effects in free recall. *Psychological Review*, 122(2), 337–363.

<https://doi.org/10.1037/a0039036>

Polyn, S. M., Norman, K. A., & Kahana, M. J. (2009a). A context maintenance and retrieval

- model of organizational processes in free recall. *Psychological Review*, 116(1), 129–156.
<https://doi.org/10.1037/a0014420>
- Polyn, S. M., Norman, K. A., & Kahana, M. J. (2009b). Task context and organization in free recall. *Neuropsychologia*, 47(11), 2158–2163.
<https://doi.org/10.1016/j.neuropsychologia.2009.02.013>
- Sederberg, P. B., Gershman, S. J., Polyn, S. M., & Norman, K. A. (2011). Human memory reconsolidation can be explained using the temporal context model. *Psychonomic Bulletin & Review*, 18(3), 455–468. <https://doi.org/10.3758/s13423-011-0086-9>
- Sederberg, P. B., Howard, M. W., & Kahana, M. J. (2008). A context-based theory of recency and contiguity in free recall. *Psychological Review*, 115(4), 893–912.
<https://doi.org/10.1037/a0013396>
- Talmi, D., Lohnas, L. J., & Daw, N. D. (2019). A retrieved context model of the emotional modulation of memory. *Psychological Review*, 126(4), 455–485.
<https://doi.org/10.1037/rev0000132>
- Varma, M. M., Zeng, S., Singh, L., Holmes, E. A., Huang, J., Chiu, M. H., & Hu, X. (2024). A systematic review and meta-analysis of experimental methods for modulating intrusive memories following lab-analogue trauma exposure in non-clinical populations. *Nature Human Behaviour*, 8(10), 1968–1987. <https://doi.org/10.1038/s41562-024-01956-y>
- Wessel, I., Krans, J., Albers, C. J., Chauhan, N., Ferreira de Sá, D. S., Hauck, A., Jaswetz, L., Michael, T., Nixon, R. D. V., Pearson, D. G., Slattery, I., Smeets, T., Takarangi, M. K. T., Willems, T. R. A., & van Schie, K. (2025). Evidence that Tetris reduces immediate but not subsequent daily intrusions of a trauma film: A multilab replication study. *Collabra: Psychology*, 11(1), 130791. <https://doi.org/10.1525/collabra.130791>
- Yonelinas, A. P., Ranganath, C., Ekstrom, A. D., & Wiltgen, B. J. (2019). A contextual binding theory of episodic memory: Systems consolidation reconsidered. *Nature Reviews Neuroscience*, 20(6), 364–375. <https://doi.org/10.1038/s41583-019-0150-4>
- Zaromb, F. M., Howard, M. W., Dolan, E. D., Sirotin, Y. B., Tully, M., Wingfield, A., & Kahana,

M. J. (2006). Temporal associations and prior-list intrusions in free recall. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 32(4), 792–804.

<https://doi.org/10.1037/0278-7393.32.4.792>

Table 2

Base parameters and theoretical roles. The parameter anchor is external to the selective-interference simulations (Dupertuys et al., 2026).

Symbol	Name	Role
β_{enc}	encoding drift rate	Context drift during study
β_{start}	start drift rate	Start-of-list or film-directed context reinstatement at retrieval onset
β_{rec}	recall drift rate	Context drift after recall
γ	feature-to-context learning rate	Learning rate for M^{FC}
α	shared support	Pre-experimental shared support in M^{CF}
δ	self support	Pre-experimental self-support in M^{CF}
ϕ_s	primacy scale	Initial boost to context-to-feature learning
ϕ_d	primacy decay	Decay rate of primacy boost
τ_c	choice sensitivity	Sharpness of retrieval competition
ϕ_{emot}	affective learning scale	Strength of affective context-to-feature learning
κ	target-monitoring scale	Reduced context updating after sampled non-film competitors
θ_s	stop scale	Baseline stopping probability
θ_r	stop growth	Growth of stopping probability across output positions

Table 3

Phase-specific encoding rates. Drift scales multiply β_{enc} and are clipped to $[0, 1]$.

Phase	Drift rate	Context-to-feature	
		learning scale	Paradigm role
Film	β_{enc}	1	Target encoding
Break	$m_{break}^{\beta} \cdot \beta_{enc}$	m_{break}^{λ}	Retention interval
Task	$m_{task}^{\beta} \cdot \beta_{enc}$	m_{task}^{λ}	Post-film task encoding
Filler	$m_{fill}^{\beta} \cdot \beta_{enc}$	m_{fill}^{λ}	Pre-recall buffer

Table 4

Scale factors and their simulation connections. Drift scales are clipped to [0, 1] after multiplication.

Scale factor	Base parameter or target	Role	Simulation connection
m_{break}^{β}	β_{enc}	Break-phase drift	Delay and context separation
$m_{\text{break}}^{\lambda}$	λ_p	Break-phase context-to-feature learning	Break-item competitor strength
$m_{\text{rem,start}}^{\beta}$	β_{start}	Start-state shift before reminder	Reminder reinstatement
m_{rem}^{β}	β_{enc}	Film-item reminder drift	Reminder reinstatement
m_{task}^{β}	β_{enc}	Task-phase drift	Target-competitor cue overlap
$m_{\text{task}}^{\lambda}$	λ_p	Task-phase context-to-feature learning	Competitor strength
m_{fill}^{β}	β_{enc}	Filler-phase drift	Post-task buffer
$m_{\text{fill}}^{\lambda}$	λ_p	Filler-phase context-to-feature learning	Filler-item competitor strength
m_{start}^{β}	β_{start}	Retrieval start reinstatement	Film-directed recall control
m^{τ_c}	τ_c	Retrieval selection sharpness	Recall control
ϕ_{emot}	affective context-to-feature learning	Affective category learning strength	Affective-category sensitivity